

2.4 Conditioning on a σ -field

Proposition 2.2. *If $\sigma(\eta) = \sigma(\eta')$, then for all $\xi \in L^1$,*

$$E(\xi|\eta) = E(\xi|\eta') \quad \text{a.s.}$$

Proof. Use Lemma 2.1 □

Definition. Let $\xi \in L^1$ and let $\mathcal{G}, \mathcal{F}$ be a σ -field. A conditional expectation of ξ given $\mathcal{G}$ is a random variable, denoted by $E(\xi|\mathcal{G})$, with the following properties:

1. $E(\xi|\mathcal{G})$ is $\mathcal{G}$ -measurable.
2. For every $A \in \mathcal{G}$, it satisfies

$$\int_A E(\xi|\mathcal{G}) dP = \int_A \xi dP.$$

Remark. For $A \in \mathcal{F}$, define the conditional probability

$$P(A|\mathcal{G}) = E(\mathbb{1}_A|\mathcal{G}).$$

Concitioning on a random variable agrees with conditioning on its generated σ -field:

$$E(\xi|\eta) = E(\xi|\sigma(\eta)) \quad \text{a.s.}$$

Theorem 2.1 (Radon-Nikodym). *Let $\mathcal{G} \subset \mathcal{F}$ be a σ -field. Then for every $\xi \in L^1$, there exists an integrable, $\mathcal{G}$ -measurable random variable ζ such that*

$$\int_A \zeta dP = \int_A \xi dP \quad \text{for all } A \in \mathcal{G}.$$

Proposition 2.3. *For every $\xi \in L^1$ and $\mathcal{G} \subset \mathcal{F}$,*

$$E(\xi|\mathcal{G}) \quad \text{exists}$$

and unique up to a.s. equality.

Exercise 2.10. Show that

$$E(\xi|\{\emptyset, \Omega\}) = E(\xi) \quad \text{a.s.}$$

Proof. Let $c = E(\xi)$ and $\mathcal{G} = \{\emptyset, \Omega\}$. Then $c \in L^1$ and it is $\mathcal{G}$ -measurable. Moreover,

$$\int_{\emptyset} c dP = 0 = \int_{\emptyset} \xi dP, \quad \int_{\Omega} c dP = c = E(\xi) = \int_{\Omega} \xi dP.$$

which implies that $E(\xi|\mathcal{G}) = c$ a.s. □

Exercise 2.11. If ξ is $\mathcal{G}$ -measurable, then $E(\xi|\mathcal{G}) = \xi$ a.s.

Proof. By assumption, ξ itself is integrable and $\mathcal{G}$ -measurable. It also satisfies

$$\int_A \xi dP = \int_A \xi dP \quad \text{for all } A \in \mathcal{G}.$$

which implies that $E(\xi|\mathcal{G}) = \xi$ a.s. □

Exercise 2.12. If $B \in \mathcal{G}$, $P(B) > 0$, then

$$E(E(\xi|\mathcal{G})|B) = E(\xi|B).$$

2.5 General Properties

Proposition 2.4. Let $\xi, \zeta \in L^1$, let $a, b \in \mathbb{R}$ and $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$.

1. (Linearity) $E(a\xi + b\zeta|\mathcal{G}) = aE(\xi|\mathcal{G}) + bE(\zeta|\mathcal{G})$ a.s.
2. $E(E(\xi|\mathcal{G})) = E(\xi)$ a.s.
3. If ξ is $\mathcal{G}$ -measurable, and $\xi\zeta \in L^1$, then

$$E(\xi\zeta|\mathcal{G}) = \xi E(\zeta|\mathcal{G}) \quad \text{a.s.}$$

4. If ξ is independent of $\mathcal{G}$, then

$$E(\xi|\mathcal{G}) = E(\xi) \quad \text{a.s.}$$

5. (Tower Property) $E(E(\xi|\mathcal{G})|\mathcal{H}) = E(\xi|\mathcal{H})$ a.s.
6. If $\xi \geq 0$ a.s., then $E(\xi|\mathcal{G}) \geq 0$ a.s.

Proof. Wirte

$$X = E(\xi|\mathcal{G}), \quad Z = E(\zeta|\mathcal{G}).$$

1. $aX + bZ$ is integrable and $\mathcal{G}$ -measurable. For all $A \in \mathcal{G}$, we have

$$\begin{aligned} \int_A (aX + bZ) dP &= a \int_A X dP + b \int_A Z dP \\ &= a \int_A \xi dP + b \int_A \zeta dP \\ &= \int_A (a\xi + b\zeta) dP. \end{aligned}$$

2. Taking $A = \Omega$ in the defining identity gives

$$E(X) = \int_{\Omega} X dP = \int_{\Omega} \xi dP = E(\xi).$$

6. Suppose $\xi \geq 0$ a.s. For

$$A_n = \{X \leq -\frac{1}{n}\} \in \mathcal{G},$$

the defining identity yields

$$0 \leq \int_{A_n} \xi dP = \int_{A_n} X dP \leq -\frac{1}{n} P(A_n).$$

Hence $P(A_n) = 0$ for all $n \geq 1$. Since

$$\{X < 0\} = \bigcup_{n \geq 1} A_n,$$

we have $X \geq 0$ a.e.

3. First assume

$$\xi \in \mathbb{1}_A \quad \text{where } A \in \mathcal{G}.$$

For every $B \in \mathcal{G}$,

$$\begin{aligned}\int_B \mathbb{1}_A Z dP &= \int_{B \cap A} Z dP \\ &= \int_{B \cap A} \zeta dP \\ &= \int_B \mathbb{1}_A \zeta dP.\end{aligned}$$

Now using linearity implies the result for simple $\mathcal{G}$ -measurable ξ . Then, use Dominated Convergence Theorem to extend the result to bounded $\mathcal{G}$ -measurable ξ . To consider the general case, that is, to show $\xi Z \in L^1$, let us define

$$H_n = (|\xi| \wedge n) \operatorname{sgn}(Z)$$

The bounded case and result in 2 give

$$\begin{aligned}E((|\xi| \wedge n)|Z|) &= E(H_n Z) \\ &= E(E(H_n \zeta | \mathcal{G})) \\ &= E(H_n \zeta) \\ &\leq E(|\xi \zeta|) < \infty.\end{aligned}$$

By Monotone Convergence Theorem, we have

$$E(|\xi Z|) \leq E(|\xi \zeta|) < \infty.$$

Now put

$$\xi_n = (-n) \vee (\xi \wedge n).$$

For every $B \in \mathcal{G}$, the bounded case gives

$$\int_B \xi_n Z dP = \int_B \xi_n \zeta dP.$$

Since $\xi_n \rightarrow \xi$ and

$$|\xi_n Z| \leq |\xi Z| \in L^1 \quad \text{and} \quad |\xi_n \zeta| \leq |\xi \zeta| \in L^1,$$

Dominated Convergence Theorem gives

$$\int_B \xi Z dP = \int_B \xi \zeta dP.$$

4. For $A \in \mathcal{G}$, independence gives

$$\begin{aligned}\int_A E(\xi) dP &= E(\xi) P(A) \\ &= E(\xi \mathbb{1}_A) \\ &= \int_A \xi dP.\end{aligned}$$

5. $E(E(\xi | \mathcal{G}) | \mathcal{H}) = E(\xi | \mathcal{H})$ is integrable and $\mathcal{H}$ -measurable. For every $A \in \mathcal{H} \subset \mathcal{G}$, we have

$$\int_A E(E(\xi | \mathcal{G}) | \mathcal{H}) dP = \int_A E(\xi | \mathcal{G}) dP = \int_A \xi dP$$

□